Week 09: October 26-30
B 3.3 Muscles & Motility (HL) and C 3.1 Integration of body systems

Subject/:الموضوع Biology **Gradeالصف : 12DP**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1	I can explain how movement occurs in animals by relating muscle contraction, motor units, antagonistic muscle action, skeletal levers, joint movement and marine mammal adaptations to the functions of movement and locomotion.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 2	I can explain and analyze how the nervous system integrates information and coordinates responses by tracing sensory input, central processing, motor output, reflex arcs, and the role of the cerebellum in movement and balance.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 3	I can explain and analyze how the endocrine system coordinates body functions by relating melatonin secretion to circadian rhythms, adrenaline secretion to vigorous activity, and hypothalamic–pituitary control to hormonal regulation.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 4	I can analyze how the nervous and enteric nervous systems regulate internal conditions by interpreting baroreceptor and	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models.	IB Biology, InThinking

	chemoreceptor input, explaining feedback control of heart rate and ventilation, and tracing the control of peristalsis.	7.2 Make learning relevant, authentic, and connected to students' backgrounds.	
Homework الواجبات			
Teacher's notes/ Reminders ملاحظات المعلم والتذكير			

Week # 09 الأسبوع#

Technology Integration دمج التكنولوجيا	dynamic sliding filament model to show the changing positions of actin and myosin during contraction and the roles of calcium ions and ATP microscopy or high-resolution sarcomere images to identify Z-lines, actin, myosin and sarcomere changes during contraction movement-analysis video or frame-by-frame playback to compare elbow flexion, breathing movements and other mechanics digital simulations or annotated videos of marine mammals to examine how body shape, flippers and tail movements contribute to propulsion and reduce drag interactive neural pathway diagrams and animations of reflex responses digital diagrams or short movement-analysis clips digital models of cardiovascular and respiratory regulation graphs showing changes in heart rate, ventilation rate, or carbon dioxide during exercise and recovery, identifying trends and explaining them using feedback control
Cross-curricular links **الروابط بين المواد الدراسية**	**Physics** • Levers, pivots and loads • Force and mechanical advantage • Friction and hydrodynamic drag • Energy transfer and movement **Physical Education** • Antagonistic muscle action • Joint mobility and range of motion • Muscle recruitment and movement control Physical Education: Balance, posture, coordination, reaction time, and movement control. Physics: Stimulus-response timing, movement, and changes in body position. Health Education: Protective reflexes and the importance of preventing injury. Computer Science: Information input, processing, and output as a simplified analogy for system integration. Mathematics: Sequencing pathways and interpreting reaction-time data, where provided.

	Physical Education: Changes in heart rate and ventilation during exercise and recovery. Chemistry: Carbon dioxide, pH, and the relationship between chemical conditions and physiological regulation. Mathematics: Graph interpretation, comparison of rates, and identification of trends during exercise and recovery. Health Education: Understanding physiological responses to physical activity and the importance of digestive function. Physics: Pressure, vessel-wall stretch, and the relationship between pressure changes and receptor input.
External Assessment Integration دمج التقييمات الخارجية	**Paper 1** Practice MCQs on: • Actin and myosin • Calcium and ATP in muscle contraction • Titin • Motor units • Antagonistic muscles • Synovial joints • Marine mammal adaptations **Paper 2** • Annotating a sarcomere diagram. • Explaining the sliding filament model. • Explaining internal and external intercostal muscle action. • Evaluating how marine mammal adaptations improve swimming efficiency. Command terms emphasized: Identify, state, describe, compare, explain, distinguish, suggest and evaluate. Paper 1: Multiple-choice questions identifying the roles of sensory neurons, motor neurons, the spinal cord, cerebral hemispheres, and cerebellum. Paper 2: Structured questions requiring students to trace a reflex arc, distinguish sensory input from motor output, explain involuntary responses, and analyze how multiple body systems contribute to a coordinated response. Paper 1: Multiple-choice questions on melatonin, circadian rhythms, adrenaline, and the hypothalamus–pituitary relationship. Paper 2: Structured questions involving melatonin concentration graphs, explanations of hormonal responses to vigorous activity, and comparisons of nervous and endocrine coordination.

	Paper 2: Data-based questions using heart-rate, ventilation, or carbon dioxide graphs; structured questions requiring explanations of feedback control, receptor input, and the neural regulation of peristalsis.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Research & Thinking (ATL)** • Linking protein interactions with whole-organism movement. • Interpreting movement diagrams and models. • Applying structure–function reasoning to unfamiliar examples. • Evaluating biological adaptations in relation to environmental demands. **Communication** • Explaining muscle contraction using precise biological terminology. • Annotating joint and sarcomere diagrams. • Constructing logical explanations from molecular to organismal levels. Research & Thinking (ATL): Analyze a biological pathway, distinguish cause from effect, and predict consequences of disrupted nervous communication. Communication: Use accurate biological vocabulary to explain a reflex arc and compare the roles of nervous-system structures. Collaboration: Work in pairs to sequence pathways, challenge explanations, and refine diagrams using evidence.
Standards المعايير	B3.3.1 Adaptations for movement as a universal feature of living organisms B3.3.2 Sliding filament model of muscle contraction B3.3.3 Role of the protein titin and antagonistic muscles in muscle relaxation B3.3.4 Structure and function of motor units in skeletal muscle B3.3.5 Roles of skeletons as anchorage for muscles and as levers B3.3.6 Movement at a synovial joint B3.3.7 Range of motion of a joint B3.3.8 Internal and external intercostal muscles as an example of antagonistic muscle action to facilitate internal body movements B3.3.9 Reasons for locomotion B3.3.10 Adaptations for swimming in marine mammals C3.1.1 System integration C3.1.2 Cells, tissues, organs and body systems as a hierarchy of subsystems that are integrated in a multicellular living organism C3.1.3 Integration of organs in animal bodies by hormonal and nervous signalling and by transport of materials and energy

	C3.1.4 The brain as a central information integration organ C3.1.5 The spinal cord as an integrating centre for unconscious processes C3.1.6 Input to the spinal cord and cerebral hemispheres through sensory neurons C3.1.7 Output from the cerebral hemispheres to muscles through motor neurons C3.1.8 Nerves as bundles of nerve fibres of both sensory and motor neurons C3.1.9 Pain reflex arcs as an example of involuntary responses with skeletal muscle as the effector C3.1.10 Role of the cerebellum in coordinating skeletal muscle contraction and balance C3.1.11 Modulation of sleep patterns by melatonin secretion as a part of circadian rhythms C3.1.12 Epinephrine (adrenaline) secretion by the adrenal glands to prepare the body for vigorous activity C3.1.13 Control of the endocrine system by the hypothalamus and pituitary gland C3.1.14 Feedback control of heart rate following sensory input from baroreceptors and chemoreceptors C3.1.15 Feedback control of ventilation rate following sensory input from chemoreceptors C3.1.16 Control of peristalsis in the digestive system by the central nervous system and enteric nervous system
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How does understanding a biological process at the molecular level improve our understanding of movement at the level of the whole organism? To what extent does understanding the individual parts of a biological system help us understand the behaviour of the whole organism? How do scientists know that a physiological response is part of a feedback mechanism rather than simply occurring at the same time as a change in the internal environment?
Success Criteria معايير النجاح (Measurable)	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: Lesson 1 MUST DOK 1 – Recall: I can identify and describe the roles of actin, myosin, calcium, ATP, titin, motor units, synovial joints and antagonistic muscles in movement. SHOULD DOK 2/3 – Apply & Analyze: I can explain how muscle contraction, motor-unit recruitment, skeletal levers and joint structure work together to produce controlled movement, including breathing movements. COULD DOK 4 – Evaluate & Create: I can evaluate how muscular, skeletal and body-shape adaptations interact to support locomotion and construct an evidence-based explanation of efficient swimming in marine mammals. Lesson 2: MUST DOK 1 – Recall: Identify the roles of sensory neurons, motor neurons, nerves, the spinal cord, cerebral hemispheres, and cerebellum.

	SHOULD DOK 2/3 – Apply & Analyze: Trace a pain reflex arc and explain how the spinal cord, brain, and skeletal muscles contribute to the response. COULD DOK 4 – Evaluate & Create: Construct and justify a systems diagram showing how nervous coordination, muscular activity, balance, and material transport contribute to an integrated response. Lesson 3: MUST DOK 1 – Recall: Identify the roles of melatonin, adrenaline, the hypothalamus, and the pituitary gland in endocrine coordination. SHOULD DOK 2/3 – Apply & Analyze: Interpret a melatonin concentration graph and explain how hormonal signalling contributes to sleep regulation or preparation for vigorous activity. COULD DOK 4 – Evaluate & Create: Construct and justify a comparative model showing how nervous and endocrine signalling integrate body functions in different situations. Lesson 4: MUST DOK 1 – Recall: Identify the functions of baroreceptors, chemoreceptors, the central nervous system, and the enteric nervous system in physiological regulation. SHOULD DOK 2/3 – Apply & Analyze: Explain how chemoreceptor input contributes to ventilation control and how neural pathways contribute to heart-rate regulation and peristalsis. COULD DOK 4 – Evaluate & Create: Construct and justify a comparative model of cardiovascular regulation, ventilation control, and peristalsis, using physiological evidence to explain the similarities and differences between their regulatory mechanisms.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Choose an item.
	المجال **Area: Marine biodiversity conservation, protection of UAE coastal ecosystems and responsible interaction with marine wildlife** **Area: Health awareness, personal safety, and responsible participation in physical activity within UAE schools and communities.** **Area: Respect for personal wellbeing, healthy routines, and the physiological needs of individuals in UAE school and community settings.**
	التطبيق **How to Implement: Students investigate how the adaptations of marine mammals support survival and locomotion in aquatic environments. They connect these adaptations to the importance of protecting marine biodiversity in UAE waters, including responsible human activities around coastal habitats. Students propose evidence-based actions that reduce disturbance to marine wildlife and support the conservation of aquatic ecosystems.**

	How to Implement: Use examples from school sports, physical activity, and injury prevention in UAE settings. Students explain how reflexes and coordinated movement help protect the body, then propose practical measures for safer participation in sports and outdoor activities. How to Implement: Students examine how sleep routines and physiological responses affect daily learning, sport, and participation in school life. They develop a realistic wellbeing recommendation for students balancing study, rest, and physical activity, recognizing that individual sleep needs and circumstances may differ.